

Conflict-Aware RAG: Multi-Stage Learning with Conflict Signals for Robust Retrieval-Augmented Generation

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Abstract

Retrieval-Augmented Generation (RAG) effectively mitigates hallucinations and knowledge gaps in Large Language Models (LLMs) for knowledge-intensive tasks by incorporating external web-based knowledge. However, when integrating diverse yet potentially conflicting web-sourced information, RAG systems are prone to knowledge conflicts that manifest as incorrect or inconsistent model behaviors, ultimately leading to unreliable responses. To address this challenge, this paper proposes Conflict-Aware RAG¹, a general training framework that leverages the model's inherent conflict-sensing capability to build a more robust RAG system via phased optimization. At the core of this framework lies ConScore, a conflict signal that quantifies the model's awareness of potential knowledge conflicts by comparing generative probabilities across distinct knowledge sources. This signal then guides both the construction of training data and a multi-stage optimization workflow: In the Supervised Fine-Tuning (SFT) stage, conflict features are employed to select representative distracting documents, laying the groundwork for core RAG capabilities; in the Direct Preference Optimization (DPO) stage, high-quality preference pairs are constructed using the conflict signal to boost the model's robustness against distracting knowledge; and in the Reranking stage, conflict confidence and information gain are integrated to synergistically optimize the collaboration mechanism between the retriever and LLM. Experiments on six knowledge-intensive question answering (QA) datasets demonstrate that Conflict-Aware RAG significantly outperforms mainstream baselines. Further ablation studies and quantitative analyses validate the method's stability and generalization, laying the foundation for robust RAG systems.

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¹Code is available at <https://github.com/ccccccchuang/Conflict-aware-RAG>



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CCS Concepts

• Computing methodologies → Natural language generation.

Keywords

Retrieval-Augmented Generation, Large Language Model, Knowledge Conflict

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1 Introduction

Large Language Models (LLMs) have recently demonstrated remarkable capabilities in language understanding and generation across a wide range of natural language processing tasks [1, 27, 37], driving rapid progress in applications such as intelligent question answering, dialogue systems, and text generation. However, the parametric knowledge within LLMs remains constrained by limited coverage and delayed updates. These limitations are particularly pronounced in knowledge-intensive tasks, where LLMs often encounter long-tail issues, factual deficiencies, and hallucinations [18, 48, 51]. Retrieval-Augmented Generation (RAG) alleviates these challenges by incorporating external web documents that are relevant to the given question. By combining external Web knowledge with the model's parametric knowledge, RAG enhances the LLMs generation process, enabling outputs that are more comprehensive, timely, and reliable [8, 22, 28].

Although RAG substantially extends the knowledge boundary of LLMs, its generation quality is still susceptible to knowledge conflicts [44] manifested during the integration of heterogeneous knowledge sources. As shown in Figure 1, knowledge conflicts primarily arise in three scenarios: 1) Inter-parametric conflict: inconsistencies within the model's internal parametric knowledge. 2) Inter-retrieval conflict: divergent or distracting information in the retrieved documents. 3) Parametric-retrieval conflict: discrepancies between retrieved external knowledge and the model's internal

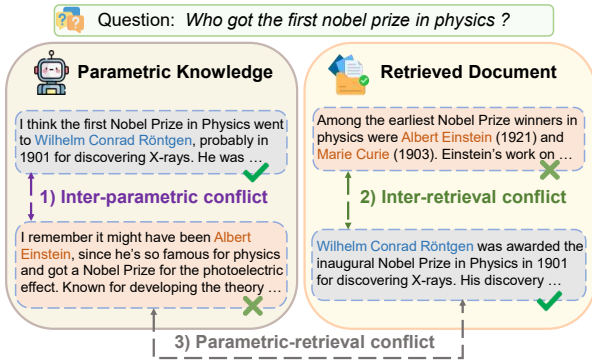


Figure 1: Conflict Scenarios in Knowledge Conflicts.

parametric knowledge. Of these, the latter two are often the primary factors limiting RAG performance. Since retrieval results may contain a mixture of supportive and distracting documents, LLMs lacking conflict awareness can be misled during knowledge integration, ultimately producing factually incorrect or unstable responses. Following RPO [46], we use the term “knowledge conflict” to refer to conflicts as perceived by the model, manifested through observable behavioral deviations or confidence shifts during knowledge integration, rather than requiring strictly verified factual contradictions between knowledge sources. Therefore, a key challenge in building robust RAG systems is empowering models to autonomously identify trustworthy knowledge while suppressing distracting information [24, 26].

To mitigate the challenges of knowledge conflicts in RAG, existing methods incorporate various alignment mechanisms during training and inference [34, 40]. Supervised Fine-tuning (SFT) enhances the LLM’s capacity to generate high-quality, document-grounded responses [2, 39]. Direct Preference Optimization (DPO) [29] trains the LLM to prefer reliable answers over distracting ones [13, 23, 35], and rerankers align retriever relevance with LLM preferences [7, 15]. However, data construction for these methods is often fragmented and lacking a unified mechanism to exploit internal signals like generation probabilities. Existing approaches rely on external LLM APIs or specialized prompts to synthesize preference pairs, which introduces high costs and distributional mismatches with real queries. This is especially problematic in the DPO stage. Since DPO is constrained by KL divergence, an excessively large initial probability gap between the “chosen” and “rejected” responses can prevent the model from adequately adjusting its preferences. Consequently, it may remain biased towards the initial higher-probability answer even if that answer is incorrect, which ultimately undermines the efficacy of preference learning [46].

Therefore, we propose **Conflict-Aware RAG**, a general framework that integrates a conflict scoring mechanism called **ConScore**, grounded in the model’s own generation likelihood disparities during both data construction and multi-stage training. ConScore quantifies the degree of conflict and shifts in knowledge confidence by comparing model perplexity gaps between correct and distracting responses across different knowledge sources, and leverages these signals to construct training samples for more accurate conflict resolution. Specifically, Conflict-Aware RAG applies ConScore across a

three-stage training pipeline: (1) SFT stage: Conflict features are extracted to select representative conflicting documents and samples, and SFT is applied to equip the LLM with basic RAG capabilities. (2) DPO stage: Conflict-based signals are used to construct high-quality preference pairs, and DPO is employed to enhance the LLM’s robustness to distracting knowledge. (3) Reranking stage: Conflict confidence and information gain guide the reranker to align with the LLM’s preferences, optimizing document ranking strategies and improving consistency and synergy between the LLM and reranker. By unifying conflict-signal modeling across stages, our framework enables conflict perception, which is primarily induced during training and manifested implicitly through the model’s generation behavior at inference, and effective handling of knowledge conflicts in RAG systems, while substantially improving overall performance. In addition, the contributions are as follows:

- We present Conflict-Aware RAG, designed to enhance the model’s capability to robustly handle conflicting or distracting information from diverse sources. It constructs training data from multiple challenging perspectives that expose knowledge conflicts, leveraging a conflict-aware three-stage training paradigm (SFT, DPO, and Reranking) to systematically improve conflict modeling and resolve multi-source conflicts more effectively.
- ConScore is introduced as a conflict-aware signal quantifying an LLM’s perception of multi-dimensional knowledge conflicts in RAG. It guides the selection of high-quality training samples for SFT and DPO and provides conflict-sensitive supervision for reranker training, enabling RAG systems to operate robustly under conflicting or distracting information.
- Extensive experiments demonstrate that Conflict-Aware RAG achieves significant performance improvements over strong baselines, validating its effectiveness on six knowledge-intensive QA datasets. Further ablation studies and quantitative analysis confirm its advantages in stability and generalization.

2 Related work

2.1 Robust RAG

RAG enhances LLMs by incorporating external knowledge [10, 22, 31]. Prior work mainly optimized retrievers [9, 14] or improved LLM adaptability [50] for robust RAG. For retriever optimization, RePlug [33] trained the retriever with LLM-generated supervision to improve relevance. RA-DIT [25] jointly trained the LLM and retriever for tighter alignment. DPA-RAG [7] aligned reranker and LLM preferences, and GainRAG [15] utilized “gain” signals to train the reranker. While these reduced preference discrepancies, retrievers might still surface non-factual content when LLMs’ preferences diverged from ground truth, especially under knowledge conflicts. In terms of task-adaptive fine-tuning applied to LLMs, methods such as Self-RAG [2] filtered noisy documents via self-reflection. Parenting [45] applied parameter decoupling and customized tuning strategies. Instruct-RAG [39] learned denoising through self-synthesis. Moreover, DPO [29] has been widely adopted for preference alignment in RAG [34, 41]. These methods trained LLMs from different perspectives using customized preference data. For instance, CR-DPO [13] self-sampled reasoning paths. RAG-DDR [23] employed DPO to jointly optimize the retrieval and generation modules. PA-RAG [40] modeled multi-perspective preferences. However, these

approaches rarely explicitly model conflict information, which is a key challenge for robust RAG. Faced with conflicting or distracting information from multiple sources, LLMs often yield unstable or inconsistent answers. Moreover, reliance on human- or expert-labeled preference data limits efficiency and adaptability. To bridge this gap, Conflict-Aware RAG generates targeted training instances by constructing conflict-aware data and leverages staged training to efficiently improve RAG performance while avoiding dependence on external expert supervision.

2.2 Knowledge Conflict

Knowledge conflicts typically arise when discrepancies between contextual information and the model’s internal parametric knowledge are manifested in the model’s generation behavior [5, 43, 44]. Previous studies have primarily attempted to mitigate this issue by explicitly improving the faithfulness of RAG systems to specific knowledge sources. Prompt-based methods [47, 49, 52] employed fixed templates to guide LLMs in explicitly distinguishing contextual knowledge from internal knowledge and reasoning about conflict origins. Decoder-based approaches [4, 16, 32, 36] leveraged token-level entropy or confidence to detect potential conflicts and dynamically weight different knowledge sources during decoding. While these methods did not require external training data, their conflict awareness showed limited generalization and robustness across diverse distributions. Other studies have aimed to improve conflict adaptation through preference training. For instance, DTA [35] jointly modeled the boundaries of parametric and retrieved knowledge, dividing data into quadrants for preference alignment. RPO [46] implicitly embedded retrieval relevance into the reward model, allowing the LLM to adaptively select among different knowledge sources. CARE [6] modeled context reliability via conflict-aware soft prompting to adaptively arbitrate between parametric and retrieved knowledge. Although these methods improved conflict handling, they generally lacked explicit modeling of LLM preferences across knowledge sources during data construction. To address this limitation, Conflict-Aware RAG constructs training data from multiple challenging perspectives that expose knowledge conflicts, leveraging intrinsic conflict signals to enhance robustness, and applies a three-stage training pipeline to systematically improve conflict modeling across all RAG modules.

3 Method: Conflict-Aware RAG

To address the challenges of data construction and training under knowledge conflicts in Retrieval-Augmented Generation (RAG), we propose the Conflict-Aware RAG framework, as illustrated in Figure 2. Section 3.1 defines the RAG task. Section 3.2 details ConScore modeling, and Section 3.3 describes conflict-aware sample construction and three-stage optimization.

3.1 Task Definition

In question answering-based RAG, the system aims to generate an answer $\hat{a}$ that is both semantically accurate and factually consistent, given a user query q and a document corpus D . The pipeline generally consists of two main components. First, a retriever R selects the top- k documents from D that are most relevant to the query q , forming the retrieved set $D_q = \{d_1, d_2, \dots, d_k\}$. Then, a generator

produces the final answer $\hat{a}$ based on D_q . This generator is typically an LLM M with strong semantic understanding and text generation capabilities. Formally, the process can be expressed as:

$$\begin{cases} D_q = R(q, D), & |D_q| = k, \\ \hat{a} = M(q, D_q). \end{cases} \quad (1)$$

3.2 Modeling Conflict Signals: Staged Construction of ConScore

By measuring confidence differences between target and distracting answers, ConScore captures interference and belief shifts across diverse knowledge sources. Subsequently, we compute the perplexity of LLMs and employ ConScore in three stages: SFT, DPO, and Reranking.

Perplexity of LLMs. Perplexity is a standard metric for quantifying the difficulty of generating a response r during the decoding process of LLMs. It is formally computed from the predicted probabilities of the generated tokens. Given a query q and a candidate response r , the perplexity (PPL) is defined as:

$$\text{PPL}(r | q) = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log P(r_i | r_{<i}, q) \right), \quad (2)$$

the lower perplexity $\text{PPL}(r | q)$ indicates that the LLM assigns higher probability to response r given query q , thereby reflecting greater confidence in this output. Furthermore, a comparison of the model’s perplexity for correct and incorrect answers under identical inputs reveals potential consistencies or conflicts between knowledge sources, forming the foundation of a conflict-aware scoring mechanism.

ConScore in SFT Stage. The objective of the SFT stage is to equip the LLM with basic RAG capabilities while improving its robustness to interference, particularly in distinguishing distracting documents. To this end, we introduce a scoring strategy, ConScores, to identify highly distracting documents. Given a query q with its reference answer a , a supporting document d^+ , and a set of n candidate distracting documents $D^- = \{d_1^-, d_2^-, \dots, d_n^-\}$, ConScores is defined to measure the interference strength of each candidate d^- . The document with the highest ConScores is then selected as the strongly distracting document for constructing the final training sample:

$$\begin{aligned} d_{\text{hard}}^- &= \arg \max_{d^- \in D^-} [\text{ConScores}(q, a, d^+, D^-)] \\ &= \arg \max_{d^- \in D^-} [\text{PPL}(a^+ | q, d^-) - \text{PPL}(a^+ | q, d^+)], \end{aligned} \quad (3)$$

where the high-quality response a^+ , generated by the LLM conditioned on the supporting document d^+ , contains the ground-truth answer a .

ConScore in DPO Stage. DPO relies on high-quality preference pairs, where the model is trained to distinguish between a “chosen” and a “rejected” response. Inspired by RPO [46], we design a conflict-aware scoring metric ConScore_D to identify borderline samples with small probability gaps between the chosen and rejected responses and low model confidence in the chosen response. Formally, for a query q with retrieved documents, a response from supporting knowledge is labeled as a “chosen” response a_c , and one from distracting knowledge as a “rejected” response a_r . Response correctness is assessed using the Exact Match (EM) metric

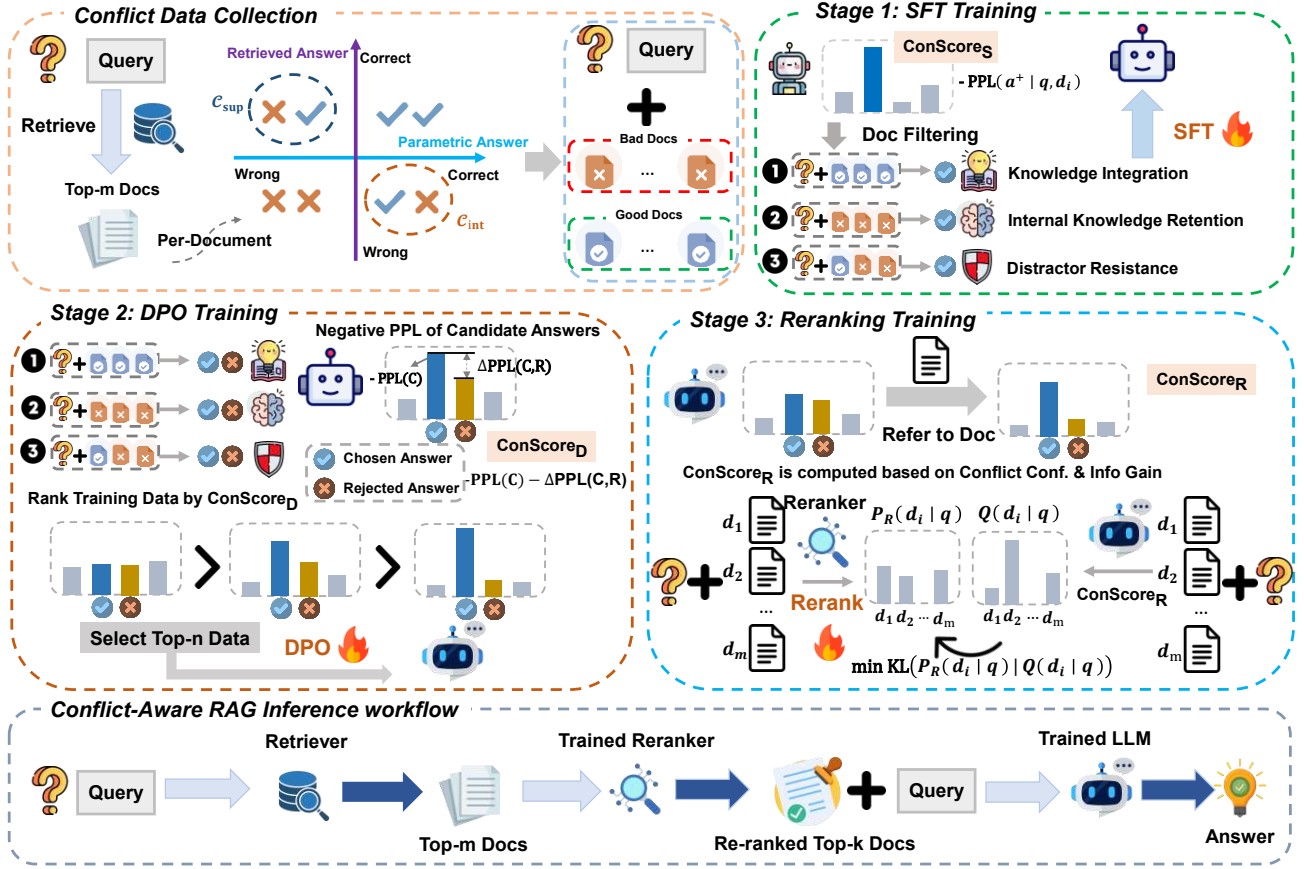


Figure 2: Illustration of the Conflict-Aware RAG framework. It depicts data selection based on the conflict signal ConScore to enhance the model’s conflict-awareness, along with the three training stages (SFT, DPO, and Reranking) and the inference workflow of Conflict-Aware RAG.

against the ground-truth answer. The conflict-aware scoring metric ConScore_D is then defined as:

$$\text{ConScore}_D(q, D_q, y_c, y_r) = (1 - \lambda) \cdot \log \widehat{\text{PPL}}(a_c | q, D_q) - \lambda \left| \log \text{PPL}(a_r | q, D_q) - \log \widehat{\text{PPL}}(a_c | q, D_q) \right|, \quad (4)$$

where $\widehat{(\cdot)}$ denotes the z-score normalization of the corresponding values over the same training dataset C , ensuring scale consistency during summation.

$$\widehat{x}_i = \frac{x_i - \mu_C}{\sigma_C}, \quad x_i \in C, \quad (5)$$

where μ_C and σ_C represent the mean and standard deviation of the corresponding values in dataset C , respectively.

ConScore in Reranking Stage. To better align retrieved documents with the LLM’s preferences in RAG, we introduce a trainable reranker optimized with preference signals derived from the LLM. Its goal is to prioritize documents that best support the model’s preferred outputs. The training dataset C is divided into a directly answerable set C^+ , where queries are correctly answered using only internal knowledge, and a document-dependent set C^- , where

correct answers require external documents. Conflict-aware supervision signals ConScore_R are then defined for both sets:

$$\text{ConScore}_R(q, d_i) = (\mu - 1) \cdot \text{CC}(\widehat{q, d_i, y^+}, y^-) - \mu \cdot \text{IG}(\widehat{q, d_i, y^+}), \quad (6)$$

where CC represents Conflict Confidence, capturing the model’s confidence in producing the ground-truth answer or resisting conflicting information from a given document, and IG stands for Information Gain, representing the confidence gain when the model is conditioned on external documents. Both quantities are z-score normalized within the training dataset, and are formally defined as follows:

$$\text{CC}(q, d_i, y^+, y^-) = \begin{cases} \log \text{PPL}(y^+ | q, d_i), & q \in C^+, \\ \frac{\log \text{PPL}(y^+ | q, d_i) - \log \text{PPL}(y^- | q, d_i)}{\log \text{PPL}(y^- | q, d_i)}, & q \in C^-, \end{cases} \quad (7)$$

$$\text{IG}(q, d_i, y^+) = \frac{\log \text{PPL}(y^+ | q, d_i) - \log \text{PPL}(y^+ | q)}{\log \text{PPL}(y^+ | q)}, \quad (8)$$

where y^+ is the ground-truth answer, and y^- denotes an incorrect answer from the model’s internal knowledge.

3.3 Data Construction and Multi-stage Training

This section presents the data construction and training pipeline of Conflict-Aware RAG, which uses the multi-stage conflict signal ConScore (Section 3.2) to integrate conflict awareness across SFT, DPO, and Reranking. In the following, we describe how ConScore guides the systematic selection of training samples and the subsequent training of the model across SFT, DPO, and Reranking stages.

Conflict Data Collection. For each query q in the dataset $\mathcal{C}$ and its corresponding set of k retrieved documents D_q , we collect responses generated by the target LLM both using only its internal parameters and using each retrieved document as context. Based on the correctness of these responses, samples are categorized into four types. (1) **Consistently Correct:** Both the LLM’s parametric answer and the document-supported answer are correct. (2) **Document Interference:** The LLM answers correctly on its own, but the provided document leads to an incorrect answer. (3) **Consistently Incorrect:** Both the LLM’s parametric answer and the retrieval-augmented answer are incorrect. (4) **Document Supplement:** The LLM fails alone, while the document provides the necessary information for a correct response. Among these, type (2) C_{int} and type (4) C_{sup} represent two typical conflict scenarios in RAG: interference from external documents and supplementation of the model’s parametric knowledge, respectively. These datasets form the focus for the construction and analysis of knowledge conflict data.

Data Construction and Training in SFT Stage. To enhance the LLM’s ability to integrate knowledge and remain robust to distracting information, we design three types of training samples. Collectively, these samples support more stable and reliable generation under multi-source knowledge conditions. 1) **Supplementary Document Knowledge Integration:** From dataset C_{sup} , we select the top- k supporting documents with highest ConScores. The LLM is trained to generate correct answers based on these documents. 2) **Internal Knowledge Preservation:** From dataset C_{int} , we select the top- k interfering documents with the highest ConScores values, indicating strong interference. The LLM learns to preserve its internal knowledge and generate correct answers despite strong interference. 3) **Document Discrimination and Robustness to Interference:** Construct a mixed-document set with one supporting document from C_{sup} and the top $k - 1$ distracting documents ranked by ConScores. The LLM is trained to identify correct information and resist misleading knowledge.

The training data constructed from the three sample types above are used to perform SFT on the LLM. The corresponding objective function optimized during SFT is defined as:

$$\mathcal{L}_{\text{SFT}} = -\mathbb{E}_{q \sim (C_{\text{sup}} \cup C_{\text{int}})} \mathbb{E}_{y \sim \pi_{\theta}(\cdot | q, D_q)} [\log p_{\theta}(y | q, D_q)], \quad (9)$$

where θ indicates the parameters of the LLM prior to SFT, and y is the expected response sequence for the corresponding sample.

Data Construction and Training in DPO Stage. In the DPO stage, the conflict datasets C_{int} and C_{sup} are resampled using the target LLM after SFT. Following the three-task design of SFT, the emphasis moves from selecting knowledge sources to constructing preference pairs at the level of generated outputs. 1) **Supplementary Document Knowledge Integration:** Randomly select k supporting documents from C_{sup} . The “chosen” response is the correct response generated by the LLM conditioned on these documents,

while the “rejected” response is the incorrect response generated using only the model’s internal knowledge. 2) **Internal Knowledge Retention:** Randomly select k distracting documents from C_{int} . The “chosen” response is the correct response based on internal LLM knowledge, and the “rejected” response is the incorrect response generated with the distracting documents. 3) **Document Discrimination and Robustness:** Combine one supporting document from C_{sup} with $k - 1$ distracting documents. The “chosen” response is the correct response generated by the LLM conditioned on the supporting document, and the “rejected” response is generated under the most distracting documents.

By constructing (“chosen”, “rejected”) preference pairs under different conflict scenarios and ranking them according to ConScore_D, we select the top-scoring subset to form the margin dataset $\mathcal{D}_{\text{margin}}$ for DPO training. In the standard DPO framework, the objective primarily maximizes the relative likelihood between “chosen” and “rejected” responses. However, under knowledge conflict scenarios, the LLM tends to reduce the generation probability of “rejected” responses rather than increase that of “chosen” responses. If the post-training probability of “chosen” responses remains low, relying solely on relative optimization may lead to insufficient discrimination when encountering other interfering knowledge. Therefore, we incorporate the SFT loss on “chosen” responses, resulting in the final DPO objective function, denoted as $\hat{\mathcal{L}}_{\text{DPO}}$:

$$\begin{aligned} \hat{\mathcal{L}}_{\text{DPO}} = \mathcal{L}_{\text{DPO}} + \alpha \mathcal{L}_{\text{SFT}} = & -\mathbb{E}_{(q, D_q, y_c, y_r) \sim \mathcal{D}_{\text{margin}}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_c | q, D_q)}{\pi_{\text{ref}}(y_c | q, D_q)} - \beta \log \frac{\pi_{\theta}(y_r | q, D_q)}{\pi_{\text{ref}}(y_r | q, D_q)} \right) \right. \\ & \left. + \alpha \log p_{\theta}(y_c | q, D_q) \right]. \end{aligned} \quad (10)$$

Training in Reranking stage. After DPO optimization, the LLM has learned conflict-aware preferences and improved robustness to distracting knowledge. Building on this model, we train a reranker to improve the sensitivity and consistency of document ranking with respect to conflict signals. For each retrieved document d_i corresponding to a query q , we first compute its conflict signal ConScore_R(q, d_i), which quantifies the document’s interference and supportive contribution to the LLM’s generation. A reranker R_{rerank} is then introduced to compute the retrieval likelihood of each document d_i given the query q :

$$P_{R_{\text{rerank}}}(d_i | q) = \frac{\exp(s(q, d_i)/\tau)}{\sum_{d \in D_q} \exp(s(q, d)/\tau)}, \quad (11)$$

where $s(q, d_i)$ is the similarity score of query q and document d_i by reranker R_{rerank} . The reranker adopts a pretrained BGE [42] model as its backbone, computing the score from the joint input of q and d_i . The parameter τ is the temperature hyperparameter that regulates the sharpness of the softmax distribution.

Since higher values of ConScore_R(q, d_i) indicate a stronger contribution to shaping the target LLM’s conflict-aware preferences, the conflict-based likelihood of each document d_i is defined as:

$$Q(d_i | q) = \frac{\exp(\text{ConScore}_R(q, d_i)/\gamma)}{\sum_{d \in D_q} \exp(\text{ConScore}_R(q, d)/\gamma)}, \quad (12)$$

where γ is an auxiliary hyperparameter controlling the softmax temperature.

We optimize the reranker R_{rerank} by minimizing the KL divergence, aligning its output distribution with the LLM’s conflict-aware distribution:

$$\mathcal{L}_{\text{rerank}} = \frac{1}{|\mathcal{B}|} \sum_{(d_i, q) \in \mathcal{B}} \text{KL}(P_{R_{\text{rerank}}}(d_i | q) \| Q(d_i | q)), \quad (13)$$

where $\mathcal{B}$ denotes the batch of training samples.

4 Experiments

4.1 Datasets and Settings

We conduct training and evaluation on six question answering (QA) datasets, including four open-domain benchmarks: Natural Questions (NQ) [21], SQuAD [30], TriviaQA [17], and WebQuestions [3], as well as two multi-hop benchmarks: 2WikiMultiHopQA (MultiHopQA) [12] and HotpotQA [11]. To comprehensively evaluate the robustness and effectiveness of LLMs, we employ Exact Match (EM) and F1 scores as evaluation metrics. EM indicates whether the generated response sufficiently covers the ground-truth answer, reflecting the model’s ability to retain and integrate correct information. F1 quantifies the token-level overlap between the generated response and the ground-truth answer, providing an overall assessment of the model’s resistance to misleading or interfering knowledge. Additional details of the datasets and prompt templates and implementation details are provided in Appendix A.

4.2 Baselines

In the experiments, we benchmark Conflict-Aware RAG against novel baselines, including: • **Vanilla RAG** [31], a basic retrieval-augmented generation method that conditions the generator on retrieved documents to produce answers. • **Self-RAG** [2], which incorporates a self-reflection mechanism during generation, allowing the model to adaptively retrieve relevant information according to task requirements and to self-assess and refine its outputs. • **DTA** [35], a method that partitions the knowledge space of RAG into four quadrants and constructs preference data to improve system reliability and trustworthiness. • **RAG-DDR-8B** [23], optimized via end-to-end differentiable data reward alignment, effectively aligning data preferences across different modules and jointly training all agents within the RAG pipeline. • **SFT on Chosen**, which performs supervised fine-tuning solely on the “chosen” responses collected from preference data. All baseline models employ identical test data, retrievers, external knowledge sources, and numbers of retrieved documents to guarantee a fair comparison.

4.3 Main Results

Experimental results are summarized in Table 1, from which the following conclusions emerge:

(1) Overall, Conflict-Aware RAG outperforms baselines in most cases across the six QA datasets in terms of both EM and F1. This shows that our framework not only generates more accurate answers but also effectively filters out distracting information, yielding concise responses. In addition, it mitigates hallucination issues under knowledge conflict and avoids verbose outputs. Notably, SFT trained solely on “chosen” responses performs worse than our

method, highlighting the advantages of incorporating DPO with preference pairs and the reranking mechanism, which together are crucial for multi-source knowledge RAG.

(2) In most cases, incorporating retrieved documents improves model performance. However, on WebQuestion and 2WikiMultiHopQA, retrieval introduces noise that degrades performance. In contrast, Conflict-Aware RAG achieves significant improvements in these settings, demonstrating its ability not only to effectively exploit useful retrieved information but also to suppress irrelevant or misleading content, thereby enhancing overall model robustness and reliability.

(3) Existing baselines such as Self-RAG and DTA rely on annotations from external LLM APIs when constructing training data, which increases both cost and dependency. In contrast, Conflict-Aware RAG uses only the target LLM’s own preference signals via a conflict-aware mechanism to filter and construct training data. Notably, without relying on external APIs, our approach outperforms these baselines across multiple datasets—underscoring how conflict signals improve data quality, robustness, and generalization.

4.4 Ablation Study

To assess the contributions of the individual training modules and the ConScore strategy in Conflict-Aware RAG, we conduct ablation studies on all datasets, with the results presented in Table 2. The key findings from our component-wise analysis are: (1) Applying SFT to Vanilla RAG substantially improves the LLM’s performance. However, removing the ConScore-based filtering of interfering documents causes a clear performance drop, suggesting that incorporating a controlled proportion of strong interference data during SFT enhances robustness. (2) Subsequent preference optimization with DPO further improves performance, demonstrating that preference data tailored to knowledge conflicts enhances the model’s sensitivity to interference and supports more accurate responses. Excluding ConScore-based data selection in this stage leads to a marked decline, underscoring the importance of high-quality filtering for effective preference learning and robust handling of conflicting information. (3) Building upon SFT and DPO, the integration of the reranker fine-tuned with conflict signals delivers additional performance gains, confirming that document preference selection guided by ConScore is crucial for optimizing overall model behavior.

4.5 Quantitative Analysis

This section presents several quantitative analyses, including the effectiveness under knowledge conflicts, the impact of preference alignment on the reranker, the effect of adjusting ConScore hyperparameters, and the stability analysis under external documents. Additional analysis is provided in Appendix B, demonstrating the analysis of training sample size based on ConScore (B.1), the efficiency analysis (B.2) and the effectiveness of external knowledge utilization (B.3).

Effectiveness under Knowledge Conflicts. To further investigate the effectiveness of Conflict-Aware RAG in knowledge conflict scenarios, we compare Vanilla RAG without fine-tuning to Conflict-Aware RAG, fine-tuned via SFT and DPO. As shown in Figure 3, Conflict-Aware RAG achieves substantial improvements in

Table 1: Performance Comparison of Conflict-Aware RAG and Various Baselines Across Six QA Datasets (EM/F1), where the best results are highlighted in bold.

Method	NQ		SQuAD		TriviaQA		WebQuestion		MultiHopQA		HotpotQA	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
LLaMA2-7B-Chat												
LLM w/o RAG	28.45	22.16	15.18	16.52	51.60	48.76	41.29	33.92	27.71	27.87	19.51	21.68
Vanilla RAG (2023) [31]	40.36	31.14	25.97	23.36	58.62	56.35	37.25	29.79	30.07	27.72	26.78	27.95
Self-RAG (2024) [2]	35.84	39.94	23.59	23.20	55.62	53.07	32.97	33.01	26.82	21.50	25.52	25.48
DTA (2025) [35]	38.17	32.27	18.53	17.56	59.98	47.14	41.58	37.01	11.88	12.18	20.41	19.50
SFT on chosen	44.96	33.59	29.13	25.42	67.38	61.02	46.06	34.41	34.18	22.14	33.10	30.45
Conflict-Aware RAG	45.93	44.95	34.73	37.51	69.60	69.97	48.57	42.78	35.31	32.85	37.93	42.86
LLaMA3-8B-Instruct												
LLM w/o RAG	28.20	30.80	16.23	21.21	58.00	60.75	37.50	37.99	23.77	24.52	20.72	25.59
Vanilla RAG (2023) [31]	43.19	42.81	27.64	30.96	63.00	63.97	35.43	35.04	29.29	18.77	28.60	32.01
Self-RAG (2024) [2]	30.72	34.75	20.76	14.12	49.38	31.15	28.25	23.36	27.14	13.73	21.94	16.67
DTA (2025) [35]	36.15	39.28	22.82	26.51	59.14	62.08	27.17	33.14	21.53	23.72	26.44	29.54
RAG-DDR (2025) [23]	45.54	46.16	30.82	35.22	68.80	70.43	41.44	40.34	33.87	26.61	32.84	36.98
SFT on chosen	46.51	46.01	30.79	35.18	69.04	69.17	47.74	42.09	46.13	25.11	36.60	39.53
Conflict-Aware RAG	51.14	49.13	36.90	40.13	73.08	71.34	52.07	42.93	50.85	23.05	43.01	42.98

Table 2: Ablation study of the components in Conflict-Aware RAG. Here, * indicates that the corresponding module did not use ConScore for data selection (i.e., data were collected randomly), while the other modules still retained the ConScore-based selection strategy.

Method	NQ		SQuAD		TriviaQA		WebQuestion		MultiHopQA		HotpotQA	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
LLaMA2-7B-Chat												
Vanilla RAG	40.36	31.14	25.97	23.36	58.62	56.35	37.25	29.79	30.07	27.72	26.78	27.95
+SFT	45.01	35.23	30.10	27.30	67.01	63.21	44.34	34.60	38.26	21.95	33.09	32.71
+SFT*	45.60	35.11	28.55	25.58	65.10	60.41	42.22	33.27	34.65	16.50	30.55	24.99
+SFT+DPO	46.01	45.64	30.78	34.07	68.60	69.00	48.03	42.19	34.76	32.23	35.18	40.26
+SFT+DPO*	44.07	43.75	30.02	33.77	63.74	64.54	40.21	38.59	31.41	30.04	32.75	37.25
Conflict-Aware RAG	45.93	44.95	34.73	37.51	69.60	69.97	48.57	42.78	35.31	32.85	37.93	42.86
LLaMA3-8B-Instruct												
Vanilla RAG	43.19	42.81	27.64	30.96	63.00	63.97	35.43	35.04	29.29	18.77	28.60	32.01
+SFT	47.04	47.03	31.26	35.43	68.20	68.83	45.57	41.76	42.95	25.76	35.10	39.62
+SFT*	46.29	47.01	29.80	33.41	65.56	65.69	40.45	39.80	27.26	23.47	29.87	32.42
+SFT+DPO	50.53	48.13	32.98	36.62	71.89	69.89	51.48	41.91	50.52	22.69	41.05	40.75
+SFT+DPO*	47.26	45.95	31.78	35.26	68.01	67.42	45.82	40.10	42.55	24.73	35.98	38.86
Conflict-Aware RAG	51.14	49.13	36.90	40.13	73.08	71.34	52.07	42.93	50.85	23.05	43.01	42.98

the “Supplemented by Document” category on the SQuAD dataset, with only a slight decline in “Interfered by Document”. These results indicate that, given valid documents, the LLM can extract accurate information to offset knowledge limitations. When faced with distracting documents, it filters irrelevant content and prioritizes reliable sources. Overall, these findings underscore that fine-tuning with conflict-aware preference data significantly enhances the LLM’s ability to handle knowledge conflicts robustly, thereby validating the intentional design and effectiveness of our framework.

Impact of Preference Alignment on the Reranker. To evaluate the reranker’s alignment with preference signals, we compare its recall and downstream performance on the NQ and SQuAD datasets pre- and post-training. As shown in Figure 4, training leads to substantial improvements in Recall@3, accompanied by consistent gains in EM and F1 scores. These findings indicate that the reranker achieves a strong preference alignment with the LLM: By leveraging ConScore-based conflict signals during training, it adapts its ranking policy to more accurately reflect the LLM’s document selection preferences.

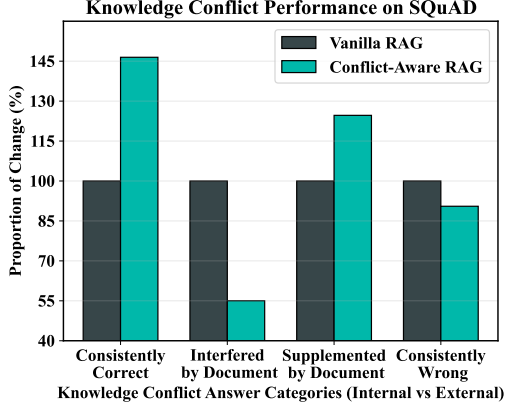


Figure 3: Performance comparison on SQuAD under knowledge conflict scenarios, using LLaMA3-8B-Instruct as the base model.

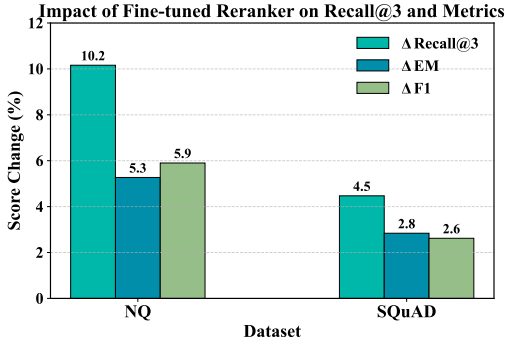


Figure 4: Changes in recall and downstream task performance of the reranker fine-tuned with preference signals on the NQ and SQuAD datasets.

Effect of adjusting ConScore Hyperparameters. To delve deeper into how key hyperparameters within the conflict signal ConScore shape outcomes, we analyze their roles in data construction. In the DPO stage, λ controls the balance between relative and absolute confidence, while in the Reranking stage, μ balances conflict confidence and information gain. Lightweight LoRA fine-tuning is applied for efficient validation. As shown in Figure 5, when λ and μ vary within the range of 0.3 to 0.7, the model performance remains largely stable, suggesting that ConScore is robust to moderate changes in these weights. To standardize our experiments, we set $\lambda = 0.5$ and $\mu = 0.5$ as default hyperparameters for Conflict-Aware RAG. In contrast, extreme cases where λ or μ is set to 0 or 1, effectively removing one component, lead to clear performance drops, confirming that all components of ConScore are essential.

Stability Analysis under External Documents. We evaluate the robustness of LLMs in the Conflict-Aware RAG framework under varying numbers of distracting documents. For each question q , distracting documents are sampled from documents associated with other queries in the same batch, with $k_{\text{int}} = 0, 1, 2, 3$. We compare Vanilla RAG, ConScore-based SFT, and Conflict-Aware RAG

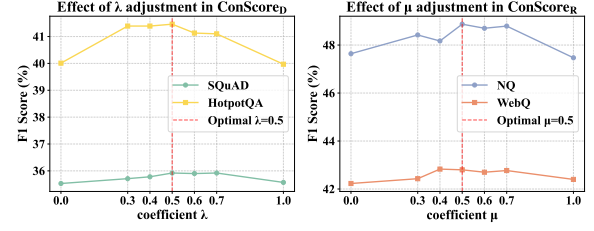


Figure 5: Sensitivity analysis of ConScore on data construction under different hyperparameter settings.

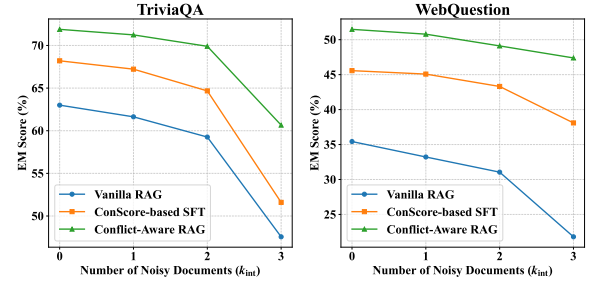


Figure 6: Effectiveness of different RAG models in resisting noisy information.

(SFT+DPO). As shown in Figure 6, Vanilla RAG’s performance drops sharply as k_{int} increases, whereas ConScore-based SFT mitigates this decline. In contrast, Conflict-Aware RAG demonstrates the highest robustness, with only minor performance loss on WebQuestions even when all retrieved documents are distractors ($k_{\text{int}} = 3$). These results show that by integrating ConScore-driven conflict signals with SFT and DPO, the model can effectively screen out distractions and zero in on relevant sources—thereby boosting overall stability.

5 Conclusions and Future Works

In this paper, we propose Conflict-Aware RAG, a general framework that integrates the conflict scoring mechanism ConScore, derived from the LLM’s own likelihood differences to guide data construction and a three-stage training pipeline—SFT, DPO, and Reranking. After training, the framework enables robust handling of conflicts across multiple knowledge sources in RAG systems, substantially improving overall performance, as confirmed by experiments on six knowledge-intensive QA datasets. Future work will focus on extending conflict-aware metrics to finer-grained and multimodal knowledge, and improving continual learning and generator-reranker coordination for better knowledge utilization.

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A More Details on Experiment Setup

A.1 Datasets

In this section, we provide a detailed description of the six QA datasets used in our experiments, including Natural Questions, SQuAD, TriviaQA, WebQuestions, 2WikiMultiHopQA, and HotpotQA. Table 3 summarizes the statistics of these datasets.

• **Natural Questions** [21] is a large-scale, high-quality English question answering dataset. Its questions are sourced from real Google search queries, and the answers are categorized into long and short forms, aiming to simulate human reading comprehension. Due to its authenticity and scale, NQ has become an important benchmark for evaluating open-domain QA systems.

• **SQuAD** [30] is a widely influential machine reading comprehension dataset based on Wikipedia articles. Due to its size and high quality, SQuAD has long served as a core benchmark for evaluating the performance of extractive QA models.

• **TriviaQA** [17] is a large-scale factual QA dataset that requires models to synthesize answers from lengthy evidence. Its complex questions and dispersed evidence present a greater challenge for reading comprehension.

• **WebQuestions** [3] is a large-scale English QA dataset with questions originating from Google search queries. Answers are typically entities or attributes from the Freebase knowledge base, designed to evaluate models' accuracy on closed-domain knowledge. Although

Table 3: Data Statistics.

Datasets	# train	# test
Natural Questions	79168	3,610
SQuAD	87599	10,570
TriviaQA	78785	11,313
WebQuestions	3778	2,032
2WikiMultiHopQA	15000	12,576
HotpotQA	90447	7,405

smaller in scale, its high quality and clear evaluation objectives make it a classic benchmark.

- **2WikiMultiHopQA** [12] is a multi-hop reasoning QA dataset, where questions require connecting information across multiple Wikipedia articles. It provides detailed reasoning path annotations, supporting research on explainable QA.

- **HotpotQA** [11] is a multi-hop reasoning QA dataset in which questions require aggregating information from multiple documents. It includes supporting sentence annotations for training and evaluating explainable QA systems.

A.2 Prompt Templates

In the SFT, DPO, and evaluation stages, we adopt a consistent prompt template to ensure uniformity in performing QA tasks:

Prompt Templates for QA Tasks

Given the following information: {Top-k document}. Answer the following question based on the given information or your internal knowledge with one or few words without the source. Question: {question}

A.3 Implementation Details

In RAG, we employ two LLMs as generators: LLaMA2-7B-Chat and LLaMA3-8B-Instruct. All datasets use the 2018 Wikipedia [20] as the external knowledge base, with e5-base-v2 [38] as the retriever. At test time, each query q is paired with $k = 3$ retrieved documents as input. For training, we perform full fine-tuning of the LLMs. During this process, we construct and filter 25k samples based on ConScore_S for the SFT stage. For DPO training, Candidate samples are ranked by ConScore_D , from which 40k samples are selected to form the margin dataset. In the Reranking stage, 22k samples labeled with ConScore_R are filtered and used. Each model is trained once on the same training set and subsequently evaluated on all six datasets to assess generalization performance.

In the SFT stage, models are trained for 3 epochs with a learning rate of $6e-6$. In the DPO stage, hyperparameters are set to $\beta = 0.3$ and $\alpha = 0.2$, and training is conducted for 1 epoch with a learning rate of $3e-6$. In the Reranking stage, we adopt the bge-reranker-base as the reranker, training with 10 retrieved documents per query q with $\tau = 1$ and $\gamma = 0.5$, for 5 epochs at a learning rate of $2e-5$. For the final evaluation, each query selects k documents from 20 candidates after reranking. The ConScore hyperparameters are set to $\lambda = 0.5$ and $\mu = 0.5$ by default. All experiments are conducted on two A800 GPUs with 80G memory.

B More Details about Quantitative Analysis

B.1 Analysis of Training Sample Size Based on ConScore

To investigate the impact of marginal sample quantity on DPO training, candidate conflict samples are ranked by ConScore_D , and subsets of the highest-scoring 10k to 60k samples are selected as training datasets for comparison. During training, we use LLaMA3-8B-Instruct as the base model and apply the lightweight LoRA fine-tuning ($r=8$, $\alpha=16$) for efficient validation, with a learning rate of $2e-5$ for 1 epoch. All other settings remain consistent with the main experiments. As shown in Figure 7, model performance improves rapidly as the number of marginal samples increases from 10k to 40k, but saturates beyond 40k. These results suggest that approximately 40k samples provide a favorable balance between model performance and dataset size, making it an effective default choice for DPO, and offering practical guidance for constructing efficient training datasets in similar alignment tasks.

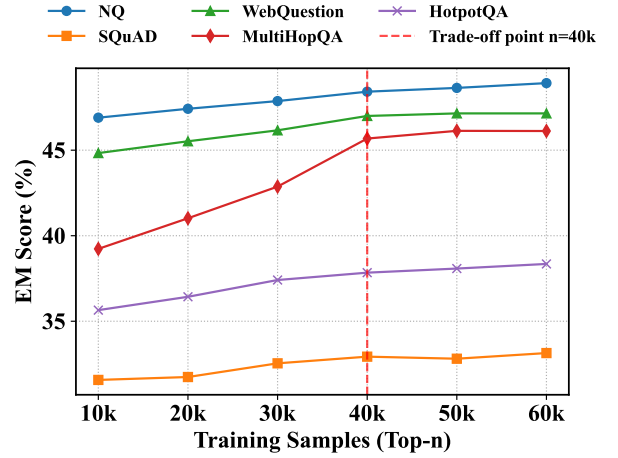


Figure 7: DPO training performance with samples selected based on ConScore_D at different scales.

B.2 Efficiency Analysis

To assess the training efficiency of our approach, we estimate the total EFLOPs (F) and training time (T) for both data construction and model training, and evaluate average EM performance (P) across six QA datasets, using the same base LLM, LLaMA3-8B-Instruct. We also indicate reliance on external expert LLM APIs (Ext. API, e.g., GPT). Following the Scaling Laws [19], training EFLOPs are approximated as $C_{\text{train}} \approx 6Nn_{\text{token}}S$, where N is the number of samples, n_{token} is the token length per sample (we use an average of 358 tokens), and S is the number of non-embedding parameters. As shown in Table 4, Conflict-Aware RAG achieves superior performance without incurring significant computational overhead. Compared to variants without ConScore computation, it attains higher performance with a tolerable increase in training time and without relying on external APIs, demonstrating both the efficiency and effectiveness of our framework.

Table 4: Efficiency analysis, Training time (T) is estimated on two NVIDIA A800 GPUs.

Method	Ext. API	#F (E)	#T (h)	#P (AVG.)
Self-RAG	Y.	15.34	42.85	30.53
DTA	Y.	2.03	5.67	32.89
RAG-DDR	N.	6.98	19.50	42.56
Conflict-Aware RAG	N.	4.21	11.76	51.89
w/o ConScore	N.	2.95	8.24	45.69

Table 5: EM Performance under Controlled External Knowledge Utilization by LLaMA3-8B-Instruct on WebQuestion and HotpotQA.

Method	WebQuestion		HotpotQA	
	have	miss	have	miss
Vanilla RAG	37.43	22.47	50.76	37.87
Conflict-Aware RAG	46.30	69.40	55.37	81.37

B.3 Effectiveness of External Knowledge Utilization

Table 5 evaluates the effectiveness of external knowledge utilization under two controlled conditions. In the “have” setting, the retrieved documents contain the golden answer, while the LLM fails to answer the question correctly using only its parametric knowledge. In

the “miss” setting, the LLM can already produce a correct answer, but the retrieved documents do not contain the golden answer and may introduce conflicting or distracting information. Experimental results show that Conflict-Aware RAG outperforms Vanilla RAG under both conditions. Notably, in the “miss” setting, where external documents are distracting, Conflict-Aware RAG maintains stable and high accuracy. These results further indicate that, through training on data constructed from model-perceived conflict signals, the model learns to selectively rely on or suppress external knowledge under varying reliability conditions, thereby improving overall system robustness.

C Limitations

Despite its effectiveness, Conflict-Aware RAG has several limitations that point to promising directions for future research.

First, in our current implementation, the prompting template explicitly encourages the model to produce concise answers, under which perplexity-based confidence shifts serve as an effective proxy for model-perceived knowledge conflict. Investigating whether and how perplexity-based conflict signals can be extended to long-form generation remains an important direction for future work.

Second, ConScore integrates multiple conflict-related components through a simple weighted summation. Although this design lacks formal guarantees of optimality, extensive empirical results demonstrate that it serves as an effective and practical heuristic for data filtering and construction. Determining the principled and adaptive ways to combine different conflict components remains a challenge.